

A B C D E F G H I J K L M N O P Q R S																								
Monte Carlo Simulation																								
Short-Term		Intermediate		Long-Term		Random 1-100		Short-Term		Intermediate		Long-Term		Investment		Short-Term		Intermediate		Long-Term		Total balance		
mean	2	5	6.25	2.5	5.5	0.3	0.55	0.15	300000	90000	165000	45000	127.9992	127.9992	127.9992	127.9992	127.9992	127.9992	127.9992	127.9992	127.9992	127.9992	127.9992	
stded	0.3	2.5	5.5	0.3	2.5	0.3	0.55	0.15	300000	90000	165000	45000	127.9992	127.9992	127.9992	127.9992	127.9992	127.9992	127.9992	127.9992	127.9992	127.9992	127.9992	
ending balance = Investment + return																								
an investor wants to invest \$300,000 in a portfolio of 3 mutual funds. The annual fund returns are normally distributed with a mean of 2.0% and Stddev of 0.3% for the short-term investment fund, a mean of 5.0% and Stddev of 2.5% for the intermediate-term fund, and a mean of 6.25% and Stddev of 5.5% for the long-term fund. An initial plan for the investment allocates 45% in the short-term, 35% in the intermediate-term, and 20% in the long-term fund.																								
Use a seed of 1, to simulate 100 trials to estimate the mean ending balance after the first year and assess the risk of this investment.																								
What if the allocation is change to 30:55:15?																								
Mean																								
Stddev																								
3131.3657																								
13121.36567																								
4.372%																								
11.000																								
30:55:15																								